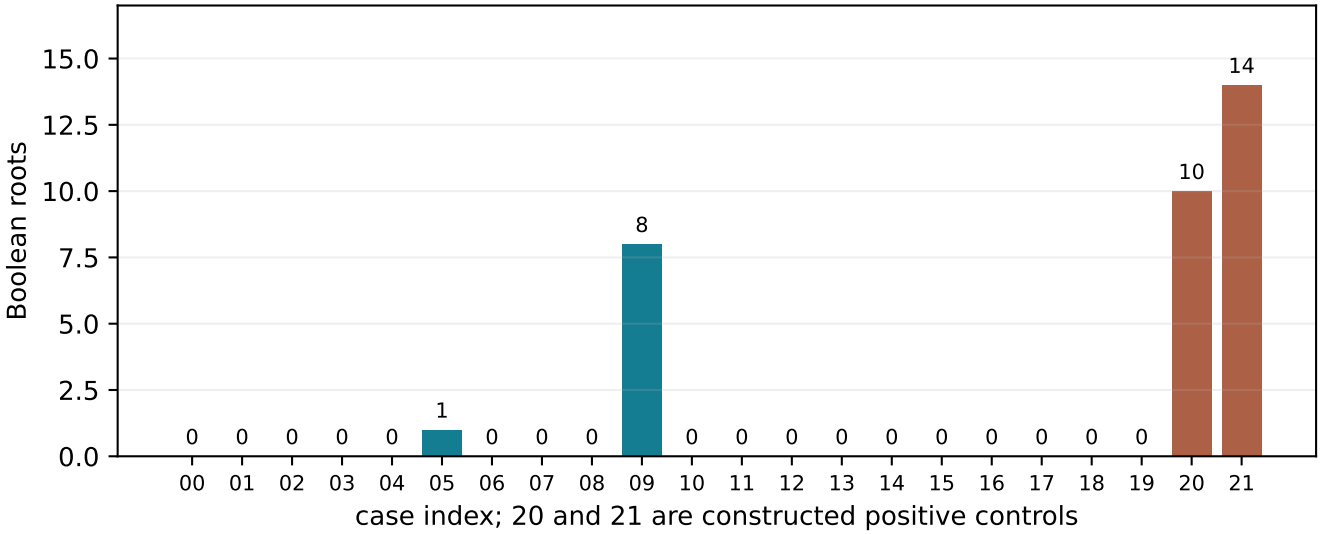
Same-input Singular correctness baseline

Local run 2026-10-25; source and protocol frozen at fa39bae96 before export. Singular 4.3.1-p3, Debian 1:4.3.1-p3+ds-2, installed in a workspace prefix from hash-checked official Debian packages. No package binary is committed.

All 44 external commands finish, with 0 failed or incomplete outputs. For each of 22 cases, both std and slimgb produce a basis with exactly the same complete Boolean zero set as independent native enumeration and native inherited-F4 root extraction. The CAS also reduces every original generator to zero against its returned basis. The strengthened Rust validator rebuilds the equations and their hashes, re-enumerates their zero set, checks the exact exported input ideal and compares every external output. It reproduces the original receipt byte for byte.



Twenty source cases come from the frozen degree-two structural panel: n=13,19, ell=2, all four basis families, replicate 0, target 0; also ell=3 geometric bases at both n. Both eliminated S4 and symmetric-e formulations retain the original equation hashes and complete root counts. These are different variable layouts; root counts across formulations need not be equal. The two additional positive algebraic controls set x_r to the first n=13 prefix basis element. They are intentionally constructed controls, with no claim to ordinary subgroup-query membership.

Singular uses characteristic 2, dp order, option(redSB), with all Boolean field equations $x_i^2 + x_i$ explicit. Each external command is single-threaded, limited to 60 seconds and 1 GiB address space; the shared x86-64 host earns L0. Resource times are retained only as diagnostics. This finite zero-set check is not a general formal Gröbner proof. A basis computation and root extraction also have different output contracts, so no speed ratio is claimed.

Magma is not licensed here; msolve, FGb and fastest-tool comparisons remain open. This check does not qualify physical performance, larger guessing cells, complete DLP costs or a rho crossover. The structural ratio figures, leaderboard and progress series remain unchanged because this run adds correctness coverage, not an algorithmic speed result.

Protocol: `../../EXTERNAL_CAS_PROTOCOL.md`. Native exporter/validator: `examples/ic_cas_correctness.rs`. Raw cases, generated scripts, package/version/host provenance, all resource and exit records, under evidence/: `runs-external-cas-20261025-01.tar.xz`. Independent receipt: `../../pipeline-receipts/external-cas-strengthened-20 replay.json`. Reproduction: `../../verify-external-cas.sh`.